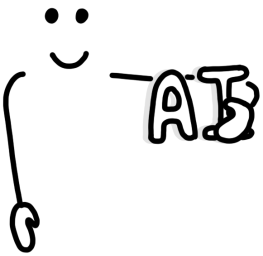
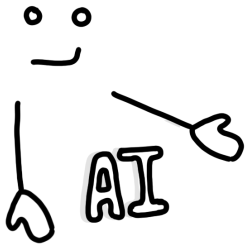
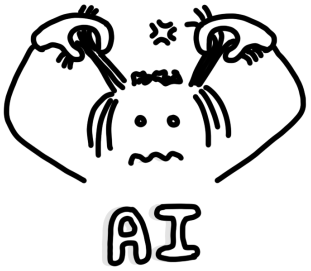
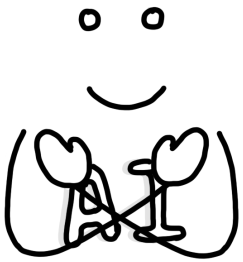
Co-existence of

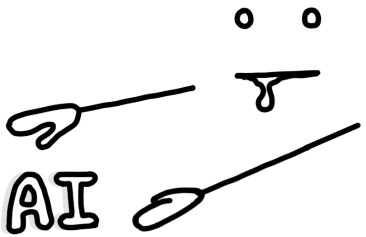
Education & AI

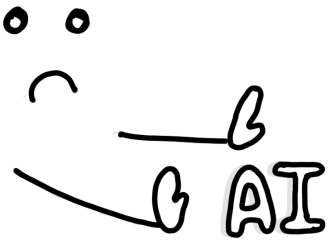












Exploration in the topic of co-existence of education and AI through the lens of systems design.

Acknowledgments

Dinesh’s feedback was sharp and exactly what was needed — I wouldn’t have made sense of these wicked problems, this complex without his guidance. And I look upto his sense of humor. Aravind’s ability to observe and make meaning of things is something I genuinely look up to.

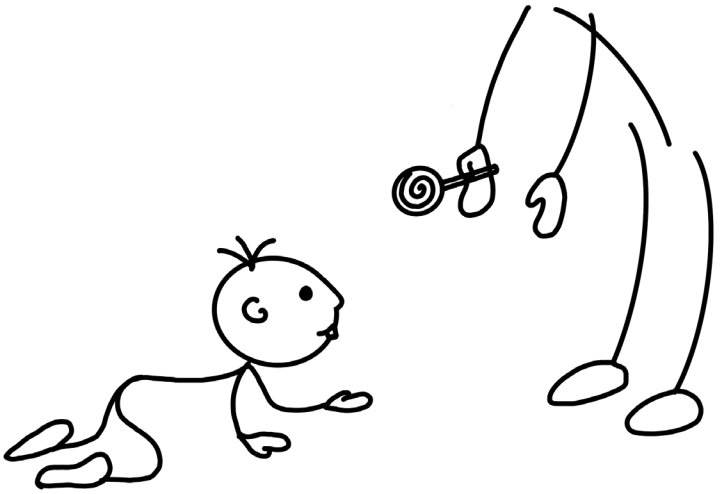
To my peers — Deetya, Anushree, Swathi, Alvin, Kashish, Arnav, Tejasvini, and Tanisha — the different perspectives and the fun we had these four months worth it.

Claude and Gemini were used to understand research sources and organize data

*thank you for the concise framing, Aravind

If Intelligence
Can Be Artificial

**Can Learning
Be Real ?**



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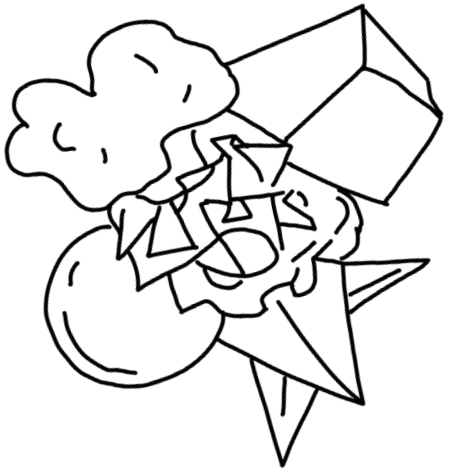
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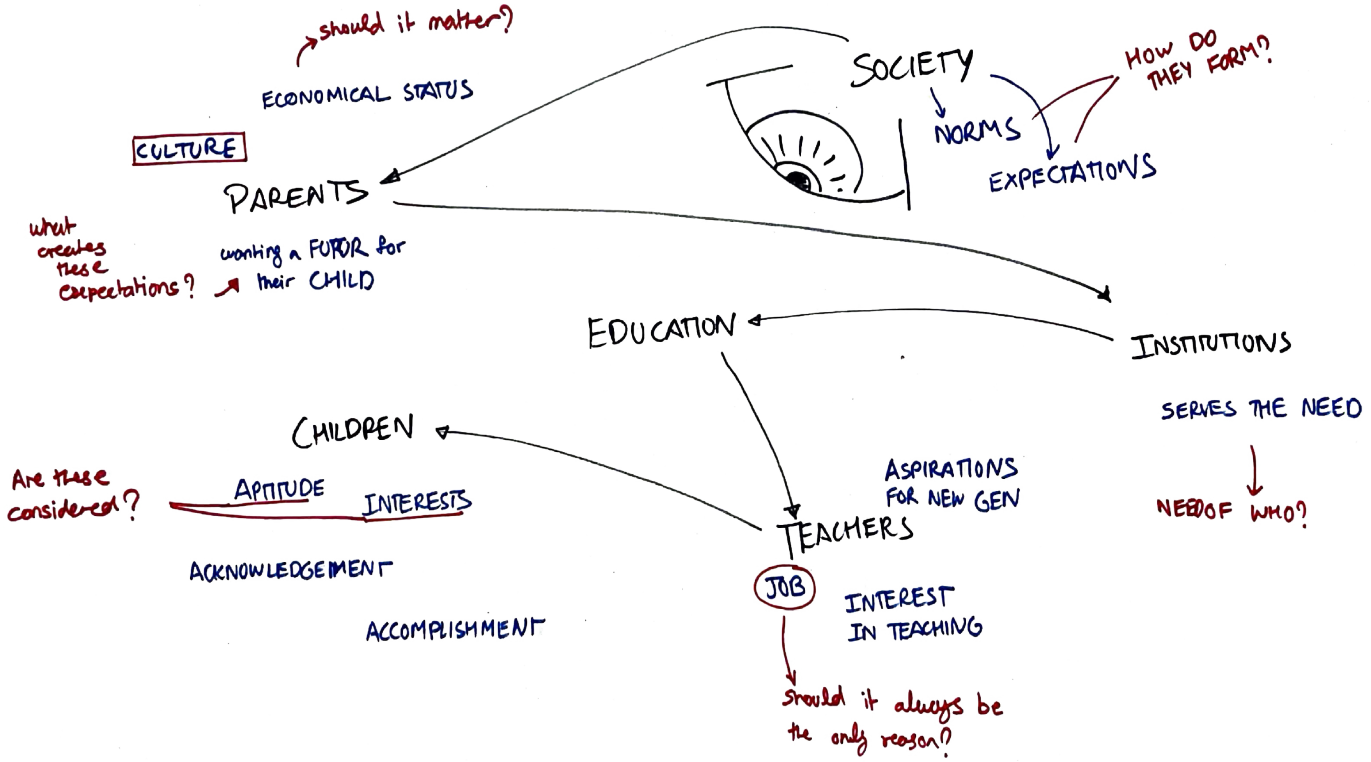
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1. Sense Making

Breaking down and understanding the topic, trying to see the connections between different elements, attempt to uncover the hidden relationships. Again stepping back and seeing it as a whole.

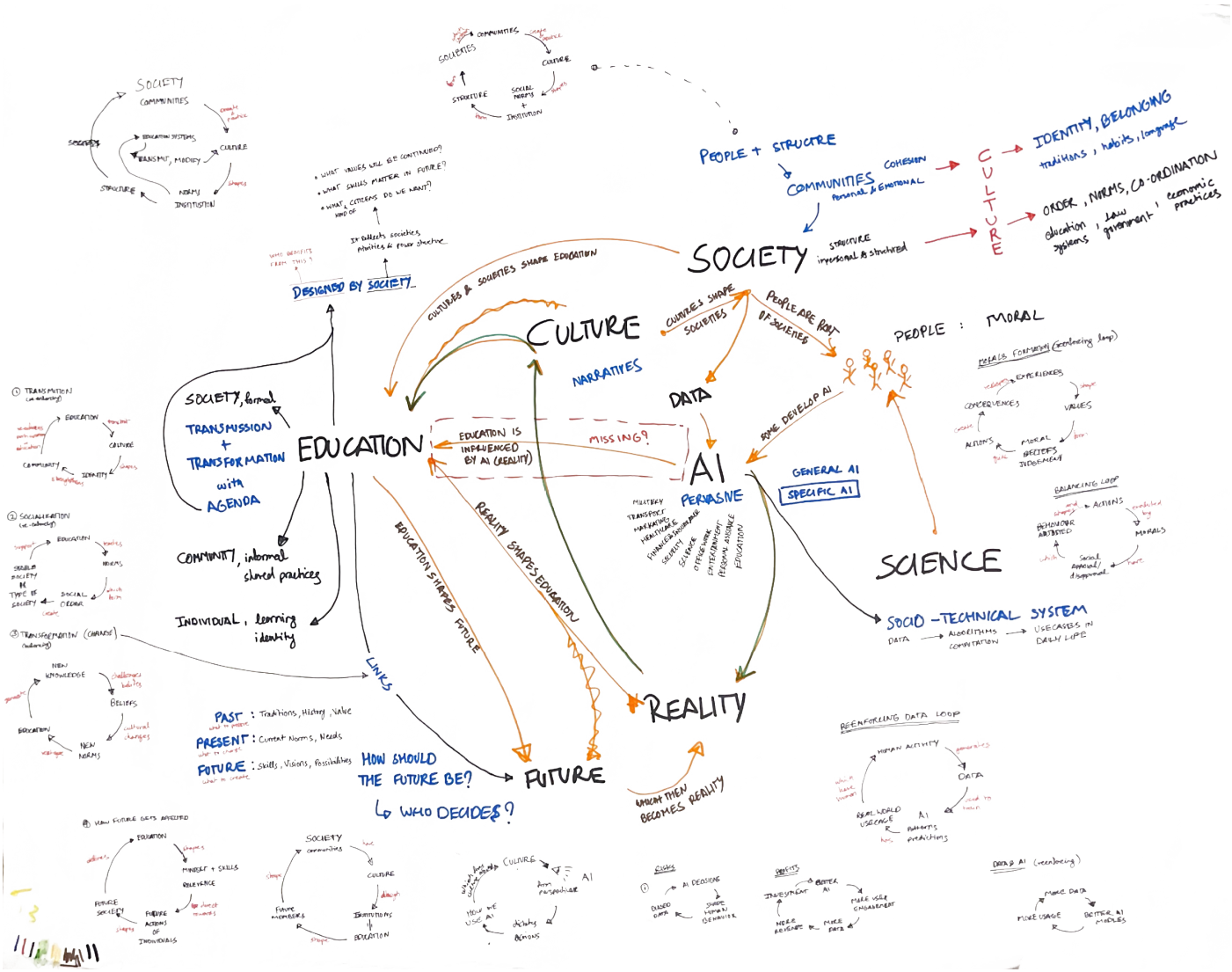
What makes education matter?



Education if forced/ becomes a norm; defines the future of masses; communities, society, country, world?

Education is not just about learning. It is tied to social status, future security & dignity

Understanding the connections, relations between social structure, future & education



Education is a controlled environment, a system developed by ___?___ to create ___?___ citizens

What factors play role when value for a learning is decided

What is learning? & When do we learn?



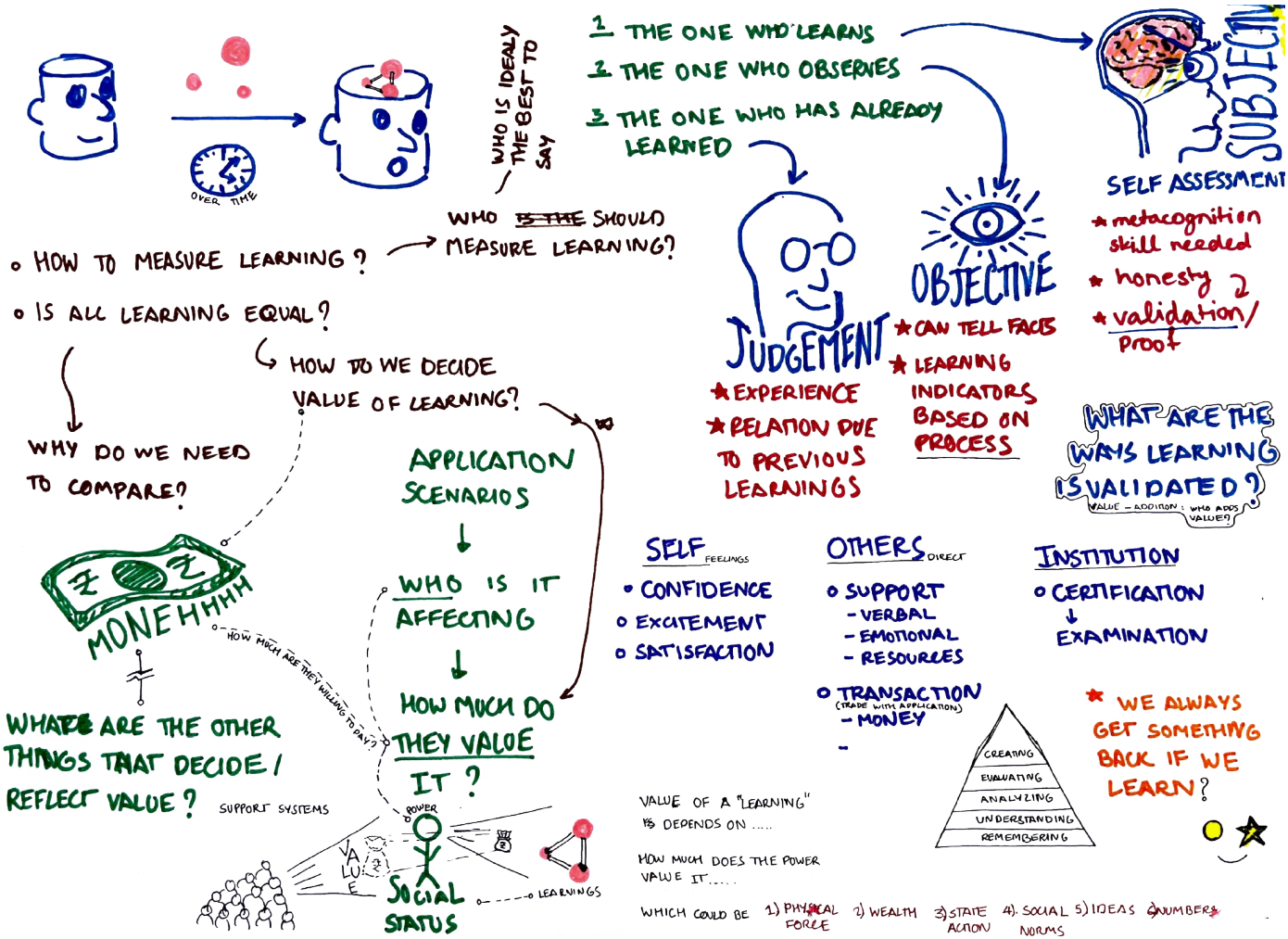
WHY & WHEN do we Learn?

Survival : out of necessity

Social : for belonging - social status

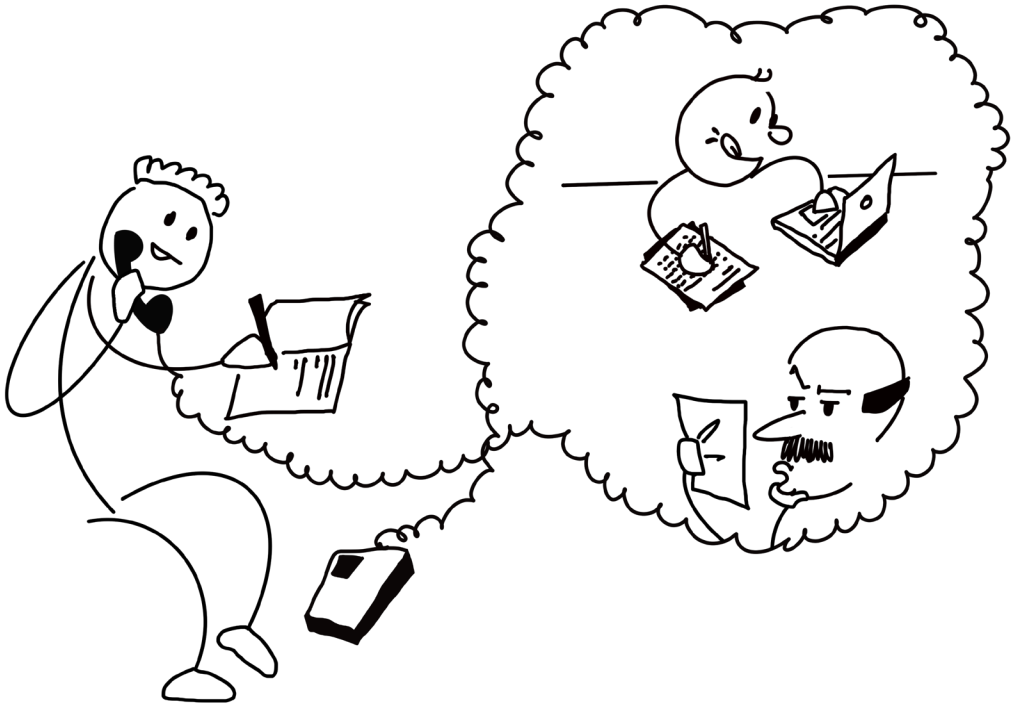
Curiosity : intrinsic motivation

How do we measure & value learning? Who should measure learning?



How to design an education system where no one feels the need to cheat

Is self assessment the best way to measure learning? Cause who else can know better than yourself?



2. Instances

These are derived from interviews and conversations with professors, teachers and students

Students say, i can find it online; what is my role then?

Science Teacher 9th-10th std

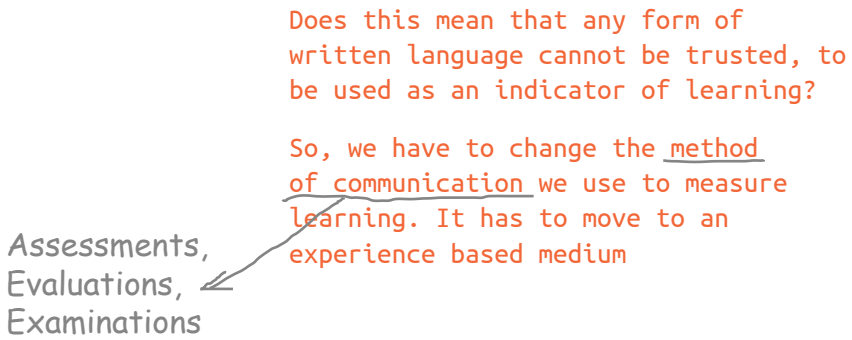
A genuine question which addresses the decrease of value for isolated retention and recall of information

But at the same time it cannot be neglected as well, as AI is very pervasive and is becoming part of our daily lives

What's the point in grading AI, then i can create questions using AI and students can answer using AI as well

-English Language Professor

A frustrated teacher who is trying to understand how to function today, where both indicators and questions can be made using AI.



Use of AI for creative writing creates new challenges

- Marathi Language Professor

An exercise which was supposed to trigger thinking, unleash the creativity can be convincingly replicated by a non - human entity; unless monitored

11th std student

Expectations for Polished work ^ (what are we valuing as a society?)

If everyone is using AI and getting away with it, while people think that work is better. Why shouldn't I use it as well?

We have to change what we value;
Policing, banning is not an option

8th std student

I used AI just for the project; i needed time for exam preparation

Shows the priorities. Even though there might be more true learning that could happen while making of a genuine project; the structure and mental model focuses on grades.

3rd yr student perusing design

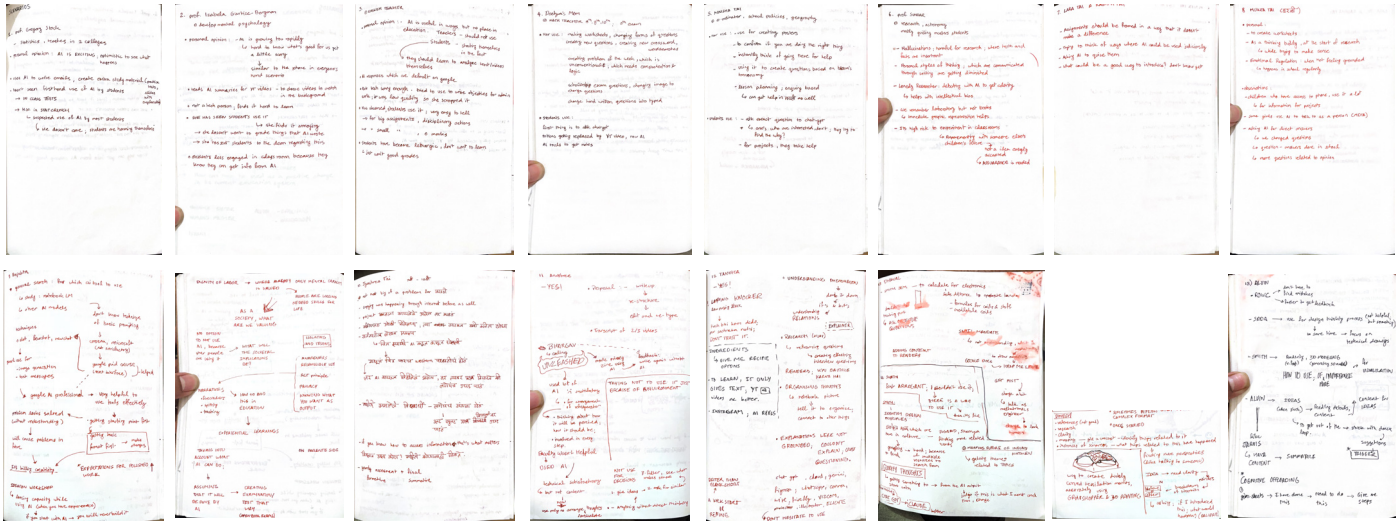
First Resort to solve any problem

It’s just too hard to stop once you start using it. You always know its there! Why would I waste my time?

Why would anyone choose to spend more time doing something when at the end the outcome is going to be valued and can be generated using AI with less time and effort

This is a feed back loop where the ‘true learning’ that needs to be valued is being skipped. And the behavior is getting re-enforced with grades

Other Interviews





3. What's the problem?

Isolating the problem by answering -

It is a problem when ...

It is not a problem when ...

It is a problem when ...

Agency is lost — students stop making their own choices, consciously or not

Learning is bypassed — intentionally or through passive habit, like procrastination

AI output is over-credited — work is validated without questioning its origin

Authorship is outsourced — writing, synthesizing, concluding, and recommending are done by AI instead of with it

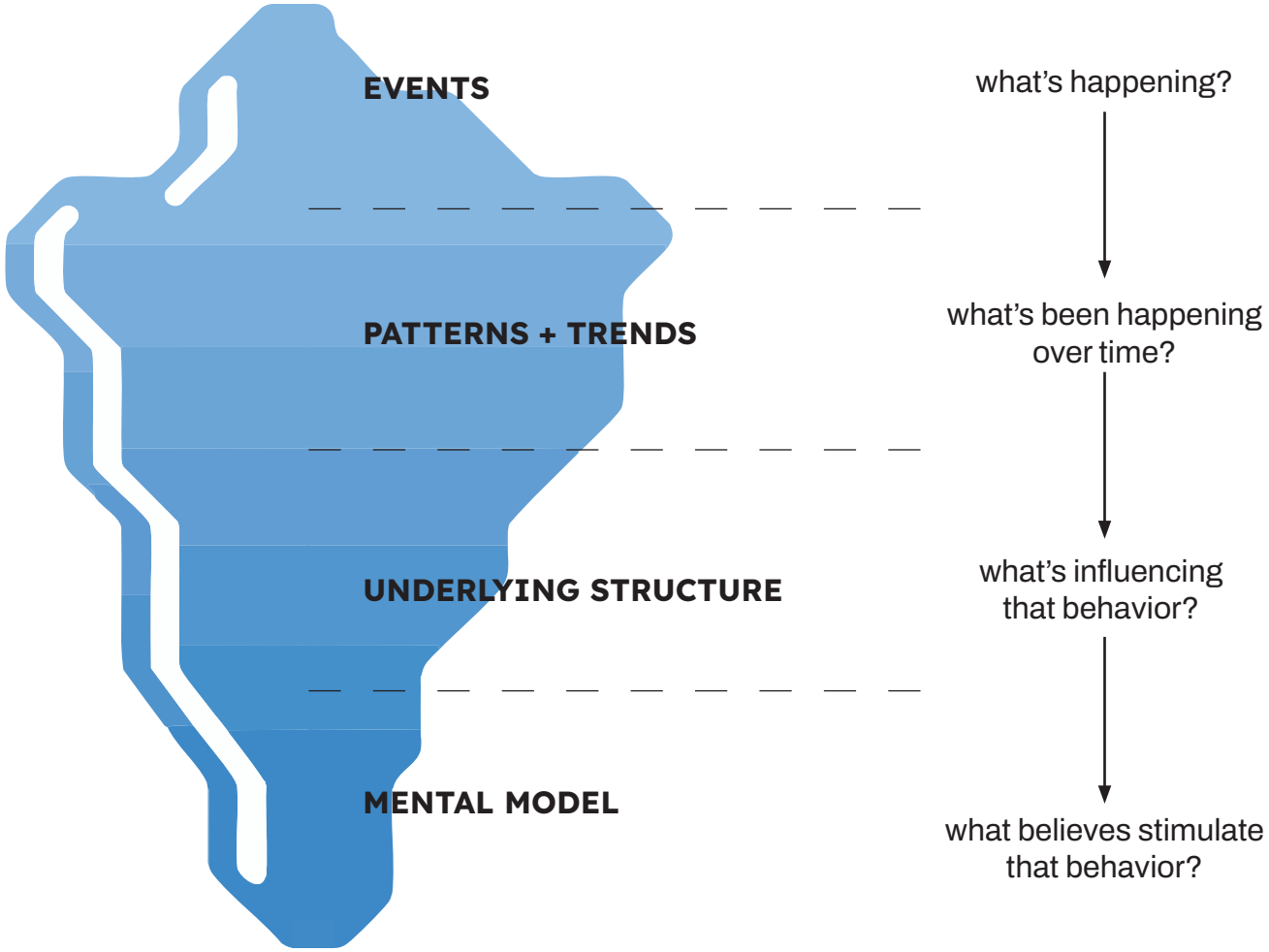
It is not a problem when ...

AI supports, not replaces — it enhances understanding without shortcutting the learning itself

Use is honestly documented — how and where AI was used is declared alongside the work

Use is visible and flagged — AI's role is identifiable, not hidden within human outcomes

*way to read iceberg model



4(a). Iceberg

Using the iceberg model to analyze, find and define the structure & mental models which are causing the surface level events

EVENTS

AI Submissions

Use of AI in Assignments

PATTERNS

AI as a first resort

Classroom Disengagement

Teachers Reactions

STRUCTURE

Grading System

Curriculum Design

Policy Regarding AI

MENTAL MODEL

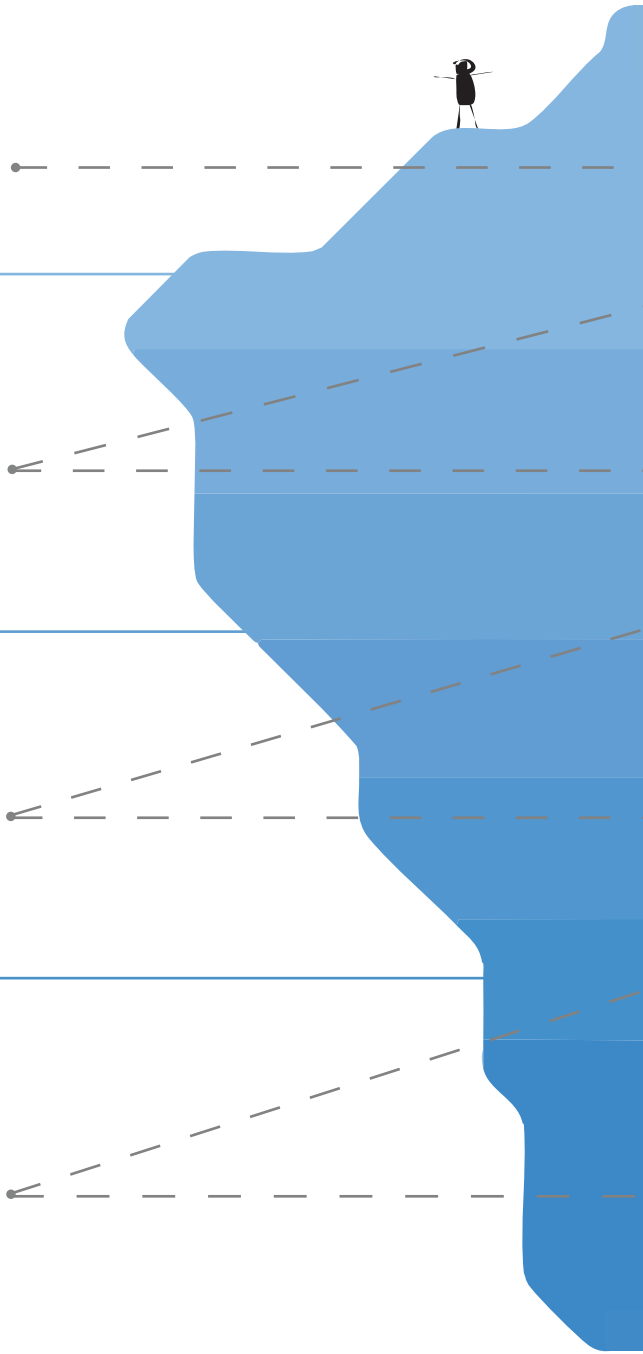
Passing = Learning

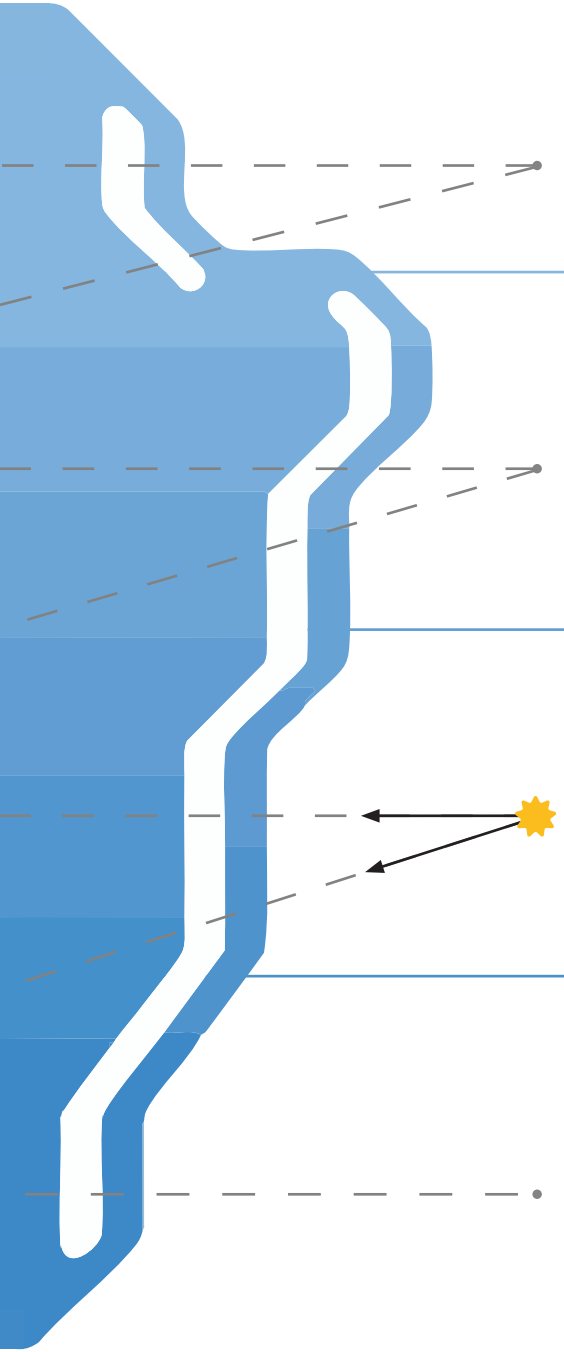
Grades = Worth

Results = Learning

Ai = Cheating / lack of efforts

Use of Language = Intelligence





Disciplinary Actions

REACTION

Efficiency in time

Avoiding effort

Lack of curiosity

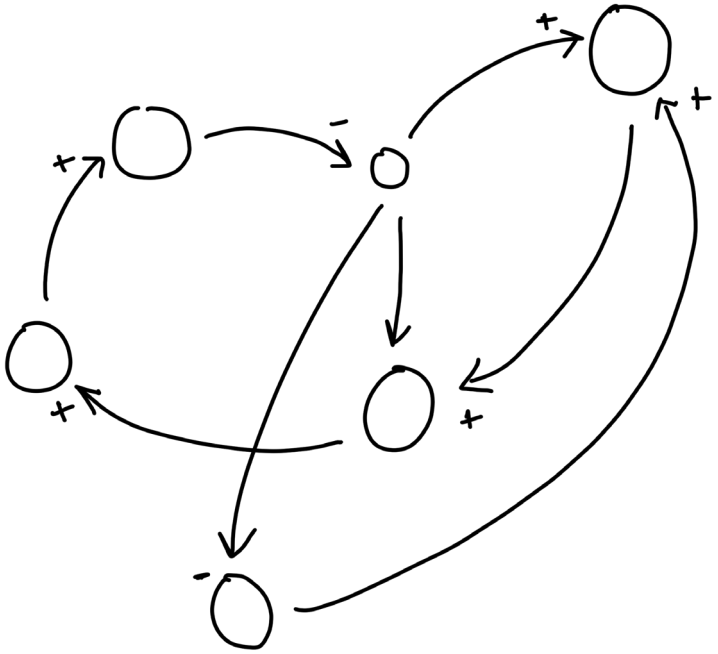
ANTICIPATE

Intervention

DESIGN

Need to change the indicators based on which we identify learning

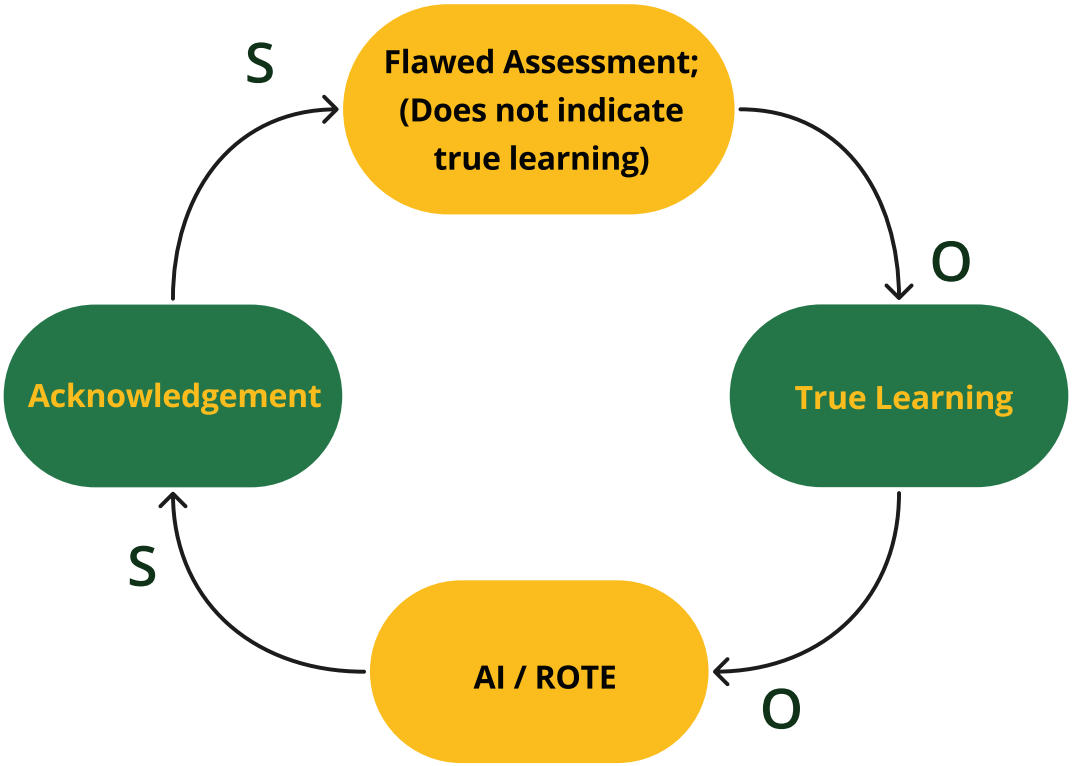
TRANSFORM



4(b). **Causal Loops**

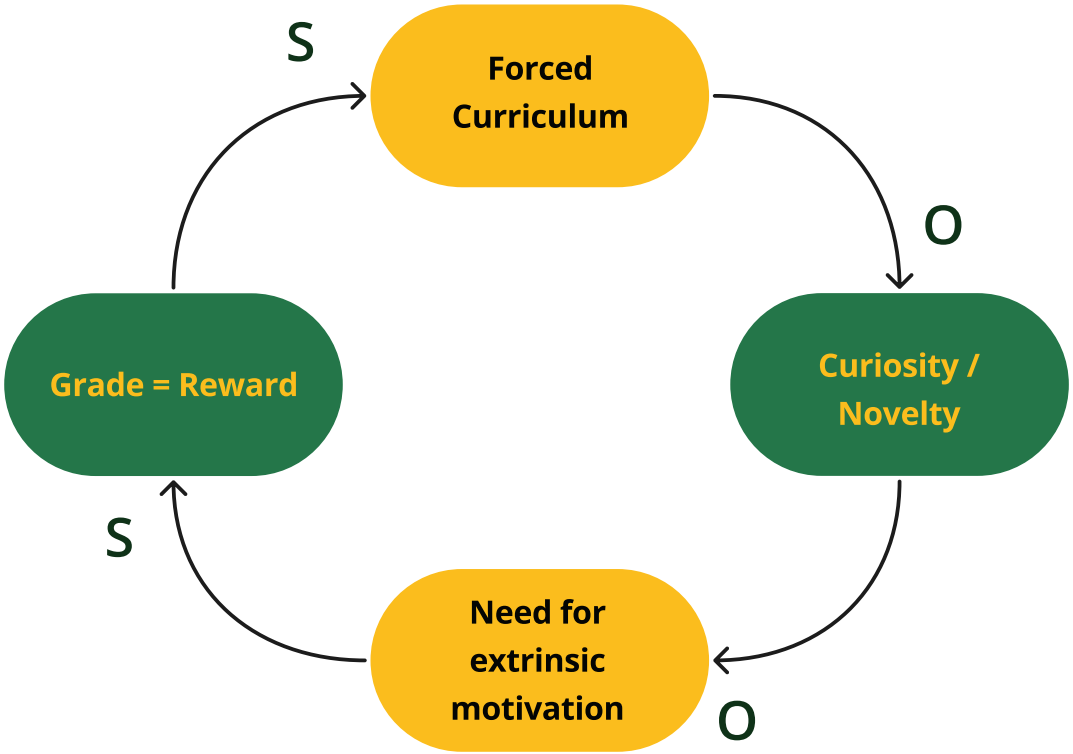
Identifying causal loops to understand and visualize the system by mapping cause-and-effect relationships

Reinforcement of AI use and Rote learning



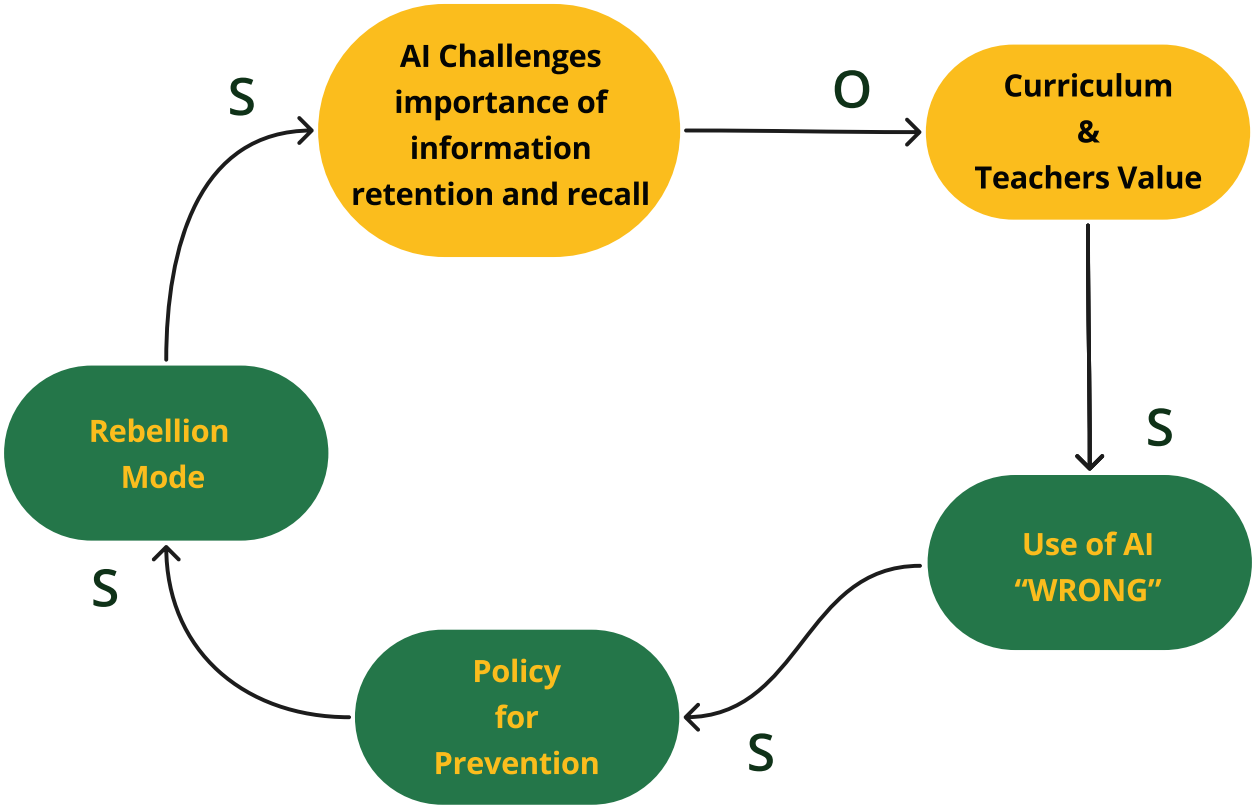
Flawed assessment does not value only true learning; it rewards rote learning, the indicators used to determine learning are easily replicable. hence, it re-enforces using AI and rote learning.

Reinforcement of importance given to grades

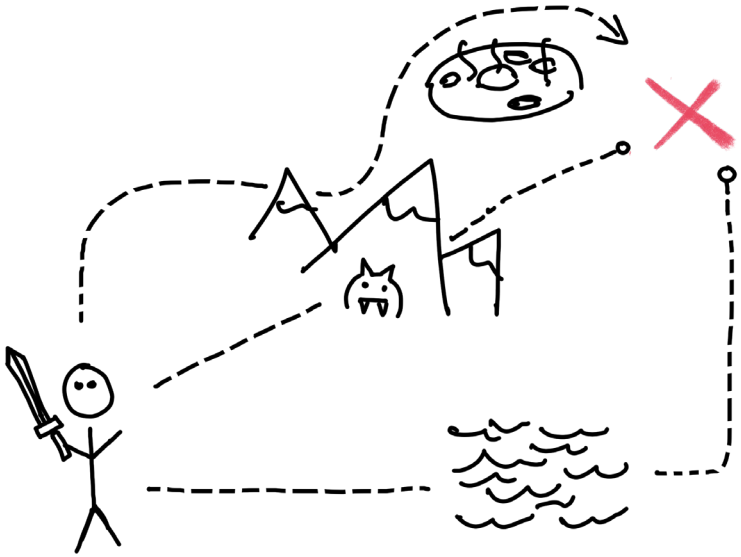


Curiosity and novelty create intrinsic motivation; but the way subjects, teaching methods are used creates a forced curriculum which creates need for extrinsic motivation. This is satisfied by giving importance and value to grades; reinforcing and adding value to grades but not learning.

Reinforcement of AI use and Rote learning

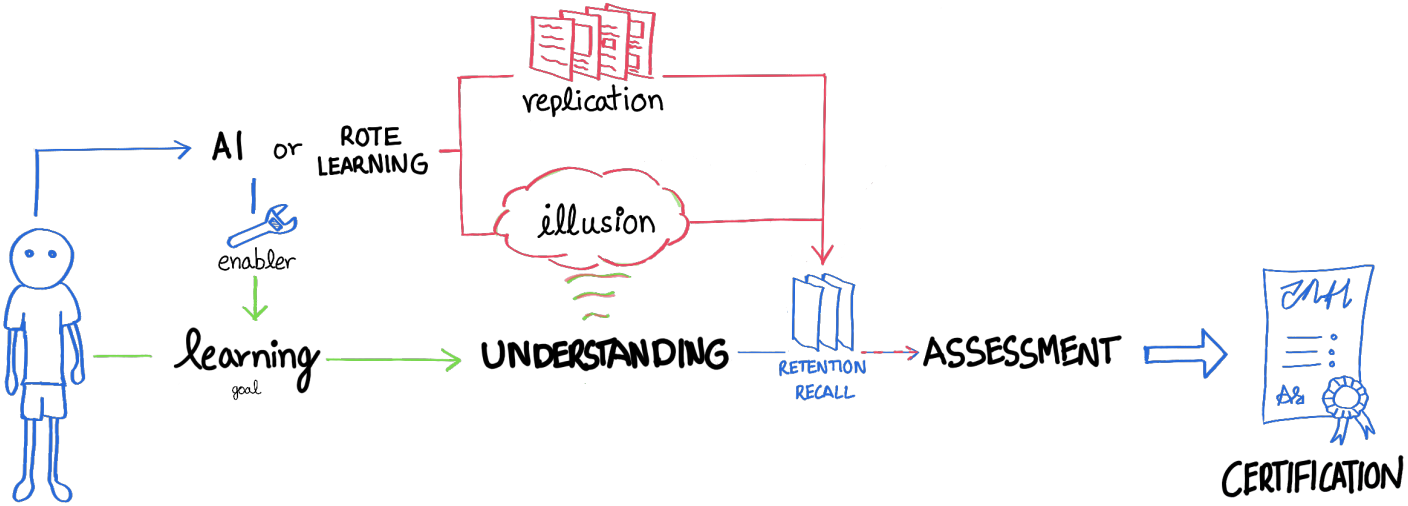


Pervasiveness and accessibility to use AI challenges the importance given to information retention and recall which is supported by the curriculum and teaching methods. Current role of teachers is based around information delivery. So AI is seen as a threat and banned to be used. This triggers a rebellion mode where students try to use AI and hide it, while teachers try to take disciplinary actions against it.



5. Structure Map

Gone through multiple iterations, structure map represents the understanding of the problem in a concise manner while illustrating the situation



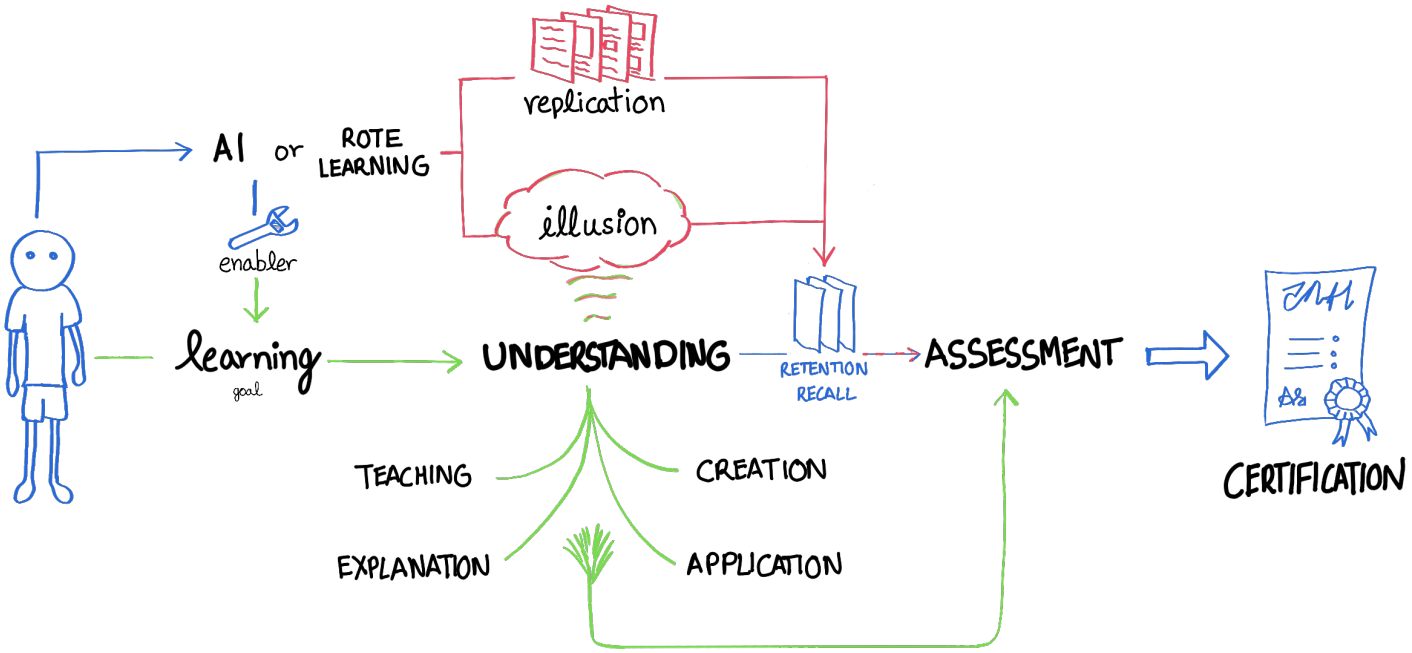
By going through the red path learning is skipped. And when only retention and recall is assessed, the red path gets approved failing to measure true learning.

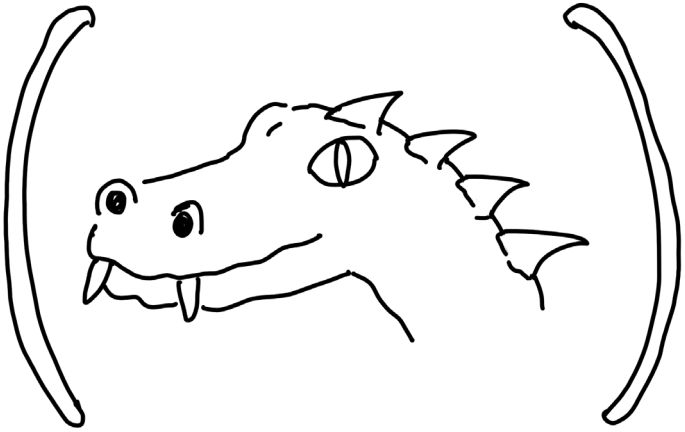
HOW TO IDENTIFY, MEASURE TRUE LEARNING

WHAT ARE THE INDICATORS OF TRUE LEARNING

Teaching/ Explanation/ Creation/ Application can be considered as indicators of true learning.

Hence, if assessment is done based on these 4 indicators, it has the potential to measure true learning. (the green path)





6. Problem Statement

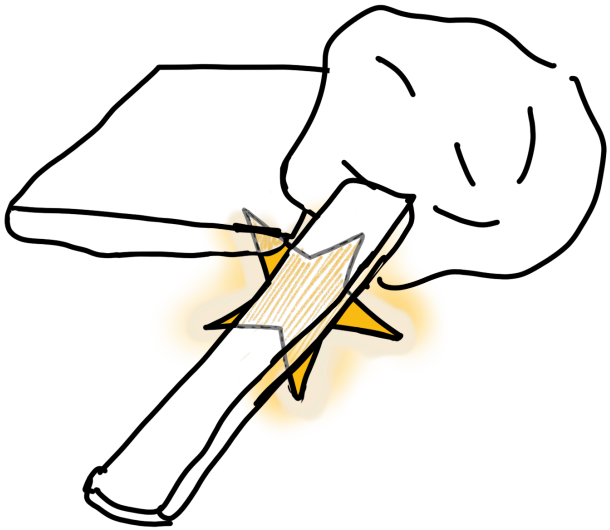
what should be considered as indicators of true learning?

in a world where,

AI and ROTE learning can replicate currently valued indicators used for assessment.

_ _

How might we change indicators used to assess so that only true learning gets measured and valued?



7(a). Insights

Assessment =! Learning

The indicators that are considered for assessment today don't necessarily reflect true learning

Curiosity & Novelty

both of these are important to create intrinsic motivation to learn, they should not be ignored

Self-assessment

is architecturally disconnected. Self-assessment exists in education as an optional, informal practice. It has no structural role in the certification chain.

Representation using Language is challenged

As any way of expressing learnings using written language cannot be trusted anymore, there is a need to assess based on different mediums, such as live experiences

7(b). Opportunities Solutions

Formalize self-assessment with comparison loops

Students assess their own competency using evidence, compare it to actual performance, and receive feedback on the gap. Not feeling-based — evidence-based. Makes assessment = learning

Change what colleges value for admission

The current system is organized around “grades = worth.” Until the goal condition changes, everything else reverts. Admission criteria need to recognize portfolios, projects, and demonstrated competency — not just exam scores.

Maker spaces affiliated with boards

Physical sites outside school where learning through creation happens — formally recognized by boards or accepted as portfolio by universities. This directly addresses the third test for true learning — creation. Shifts the question from “did you absorb the content?” to “did you make something with it?” Affiliation with formal structures prevents it from being dismissed as extracurricular.

Nested task redesign tool for teachers

A one-page card or guide: take any rote task, identify its level, design a second task that overlaps the content one level higher — teach it, explain it, apply it. No institutional approval needed. One teacher, one class, next week.

Formalize reflective journaling

A structured, recognized channel for students to receive information about their own learning — not as homework, but as a formal practice. Makes the invisible process visible. This creates new information flow which is extremely useful for learning

Reformat textbooks for group-based learning

Shift the medium from solo absorption to dialogue, teaching, and presentation. Nested tasks where feedback from one task is usable in the next. This naturally triggers meta cognition, fulfills one of the identified indicators of true learning (teaching someone else).

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